5. Python for Data Analysis: Data Wrangling with Pandas, NumPy & Jupyter — *Wes McKinney*, Shroff Publishers & Distributors (Indian edition of O'Reilly Media)
6. Machine Learning in Data Science Using Python — *Dr. R. Nageswara Rao*, Wiley India / Dreamtech Press India Pvt Ltd (Indian publisher)
7. Hands-On Data Analysis and Visualization with Pandas — *Purna Chander Rao & Kathula*, BPB Publications (Indian publisher)
8. NumPy, Pandas, and Scikit-learn Masterclass — *Gary Hutson*, BPB Publications (Indian publisher)
9. Advanced Data Analytics Using Python: With Machine Learning, Deep Learning & NLP Examples — *Sayan Mukhopadhyay*, Apress (Springer Nature)
10. Python Data Analysis – Third Edition — *Avinash Navlani & Ivan Idris*, Packt Publishing (international technical publisher)

SUBJECT NAME: Research Methodology **Credit: 3**
SUBJECT CODE: BCAC702

COURSE OBJECTIVE:
This course aims to provide learners with a clear understanding of the fundamental principles and systematic procedures involved in conducting research. Students will learn how to identify and define research problems, formulate objectives and hypotheses, and design appropriate research plans. The course develops the ability to carry out effective literature reviews, recognize research gaps, and select suitable sampling and data collection methods such as surveys, interviews, and observations. Learners will acquire basic knowledge of statistical tools required for organizing, analyzing, and interpreting data, including estimation, correlation, regression, and hypothesis testing. Emphasis is given to data presentation through tables, charts, and graphs for meaningful interpretation of results. The course also focuses on developing skills for writing research reports, preparing proposals, and following standard referencing styles. Ethical practices in research, including avoidance of plagiarism and responsible data handling, are highlighted. By the end, students will be capable of planning, conducting, analyzing, and presenting research effectively in academic and professional settings

COURSE OUTCOME:

CO No.	Course Outcome (CO)
CO1	Identify and define research problems, formulate objectives and hypotheses, and design an appropriate research plan.
CO2	Conduct effective literature reviews, identify research gaps, and select suitable sampling and data collection methods.
CO3	Apply basic statistical tools such as estimation, correlation, regression, and hypothesis testing to analyze and interpret research data.

CO No.	Course Outcome (CO)
CO4	Prepare structured research reports and proposals using standard referencing styles while adhering to ethical research practices.

DETAILED SYLLABUS:

Module No.	Detailed Course Structure	Hours	Marks (100)
M 1	**Foundations of Research** • Meaning, objectives, and types of research (basic, applied, exploratory, descriptive) • Research process and characteristics of good research • Identifying and defining a research problem • Formulation of research questions and objectives • Variables in research and research motivation	10	15
M 2	**Literature Review and Research Design** • Importance of literature review • Sources of literature: journals, conference papers, reports, online databases • Research gap identification • Types of research design: experimental, descriptive, analytical • Developing hypotheses and conceptual framework	10	15
M 3	**Sampling Techniques and Data Collection Methods** • Census vs sampling • Probability and non-probability sampling methods • Primary and secondary data collection • Methods: surveys, observation, interview, questionnaire • Designing questionnaires and schedules	10	20
M4	**Statistical Tools for Research** • Classification and tabulation of data • Measures of central tendency and dispersion • Correlation and regression basics • Estimation of population parameters • Hypothesis testing: t-test, chi-square test, ANOVA (concept)	12	20
M 5	**Data Analysis, Interpretation and Presentation** • Data coding and data processing • Use of charts, tables, and graphs for presentation • Interpretation of statistical results • Drawing conclusions and recommendations • Use of software tools for analysis (overview)	10	15
M 6	**Research Reporting and Ethics** • Structure of research report / thesis / paper • Writing abstract, introduction, methodology, results, conclusion • Referencing styles (APA/MLA/IEEE) • Plagiarism and research ethics • Preparing research proposal and presentation	8	15

Reference Books:

1. Research Methodology: Methods and Techniques — C. R. Kothari — New Age International Publishers
2. Research Methodology: A Step-by-Step Guide for Beginners — Ranjit Kumar — Sage Publications India

3. Research Methods for Business: A Skill-Building Approach — Uma Sekaran & Roger Bougie — Wiley India
4. Fundamentals of Research Methodology and Statistics — Yogesh Kumar Singh — New Age International Publishers
5. Business Research Methods — Donald R. Cooper & Pamela S. Schindler — McGraw-Hill Education (India)
6. Social Science Research: Principles, Methods & Practices — Anol Bhattacherjee — University of Florida Press (widely used in India)
7. Research Design: Qualitative, Quantitative, and Mixed Methods Approaches — John W. Creswell & J. David Creswell — Sage Publications (International edition often available in India)

Subject Name: Research Methodology Lab **Credit: 2**
Subject Code: BCAC792

Course Objective:
The course aims to develop a clear understanding of the principles, processes, and techniques involved in conducting systematic research. Learners will gain the ability to identify research problems, formulate objectives and hypotheses, and design appropriate research plans. The course emphasizes effective literature review, suitable sampling, and data collection methods. Students will learn to apply basic statistical tools for organizing, analyzing, and interpreting data. It also focuses on developing skills in research report writing, proper referencing, and maintaining ethical standards. By the end, students will be able to plan, execute, and present research studies confidently in academic and professional contexts.

What to be done:
Students should begin by identifying a relevant research problem and framing clear objectives and hypotheses. They must conduct a thorough literature review using reliable sources to identify research gaps. Based on the problem, an appropriate research design and sampling method should be selected. Learners need to prepare data collection tools such as questionnaires or interview schedules and gather primary or secondary data systematically. The collected data should be organized, coded, and presented using tables, charts, and graphs. Basic statistical techniques such as estimation, correlation, regression, and hypothesis testing should be applied to analyze the data. Students must interpret the results carefully and draw meaningful conclusions. Finally, they should prepare a well-structured research report or proposal following standard referencing styles and ethical practices, ensuring originality and clarity in presentation.

Experiment No.	Title of Experiment	Learning Outcome
1	Introduction to Research and Identification of Research Problem	Understand different types of research and formulate research problems.
2	Literature Review using Digital Libraries	Search and review research papers using databases such as Google Scholar, IEEE Xplore, Springer, and ACM Digital Library.
3	Research Objectives, Questions and Hypothesis Formulation	Develop research objectives, research questions, and hypotheses.
4	Research Design and Sampling Techniques	Learn qualitative, quantitative, and mixed research methods along with sampling techniques.
5	Designing Questionnaire using Google Forms	Create structured questionnaires for data collection.
6	Data Collection and Data Cleaning	Collect primary/secondary data and perform preprocessing.
7	Statistical Analysis using MS Excel/SPSS/R	Compute descriptive statistics such as mean, median, mode, standard deviation, and variance.
8	Data Visualization	Generate charts, graphs, and dashboards for data interpretation.
9	Hypothesis Testing	Perform t-test, chi-square test, ANOVA, or correlation analysis based on sample data.
10	Citation and Reference Management	Use tools such as Zotero or Mendeley for citations and bibliography in IEEE/APA formats.
11	Plagiarism Checking and Ethical Research Practices	Check similarity using plagiarism detection tools and understand research ethics.

Experiment No.	Title of Experiment	Learning Outcome
12	Research Paper Writing using IEEE/APA Format	Prepare a complete research paper including Abstract, Introduction, Methodology, Results, Conclusion, and References.

CO	Description
CO1	Identify research problems and formulate research objectives and hypotheses.
CO2	Conduct literature surveys using standard digital repositories and reference management tools.
CO3	Design research methodologies, collect data, and apply appropriate statistical techniques.
CO4	Interpret research findings, maintain research ethics, and prepare technical reports or research papers following standard publication formats.